NAME - KRISH CHANDRASHEKHAR PACHODE ROLL NO.- SY-B-18 BATCH - AIDS(SY-B)

AIM -

To study and implement version control using Git and GitHub by creating a local repository, performing commit operations, and pushing files to a remote repository.

THEORY -

1. What is the difference between Git and GitHub?

Git is a distributed version control system used to track changes in files and manage source code history.

GitHub is a cloud-based platform that hosts Git repositories and allows collaboration among developers.

Difference:

- Git → Tool/software for version control
- GitHub → Online platform for storing and sharing repositories

---

2. Explain the difference between:

git add

- Moves changes/files to the **staging area**
- Prepares files before committing

git commit

- Saves the staged changes permanently in the repository
- Creates a snapshot of the project

---

3. What is the staging area in Git?

The staging area is an intermediate space where changes are stored before committing. It allows users to review and select specific changes before saving them permanently.

---

4. What happens if you modify a file after committing it?

If a file is modified after committing:

- Git marks it as **modified**
- You need to run `git add` again to stage the changes
- Then use `git commit` to save the updated version

---

5. Difference between:

git push

- Uploads local commits to a remote repository (GitHub)

git pull

- Fetches and updates local repository with changes from remote repository

---

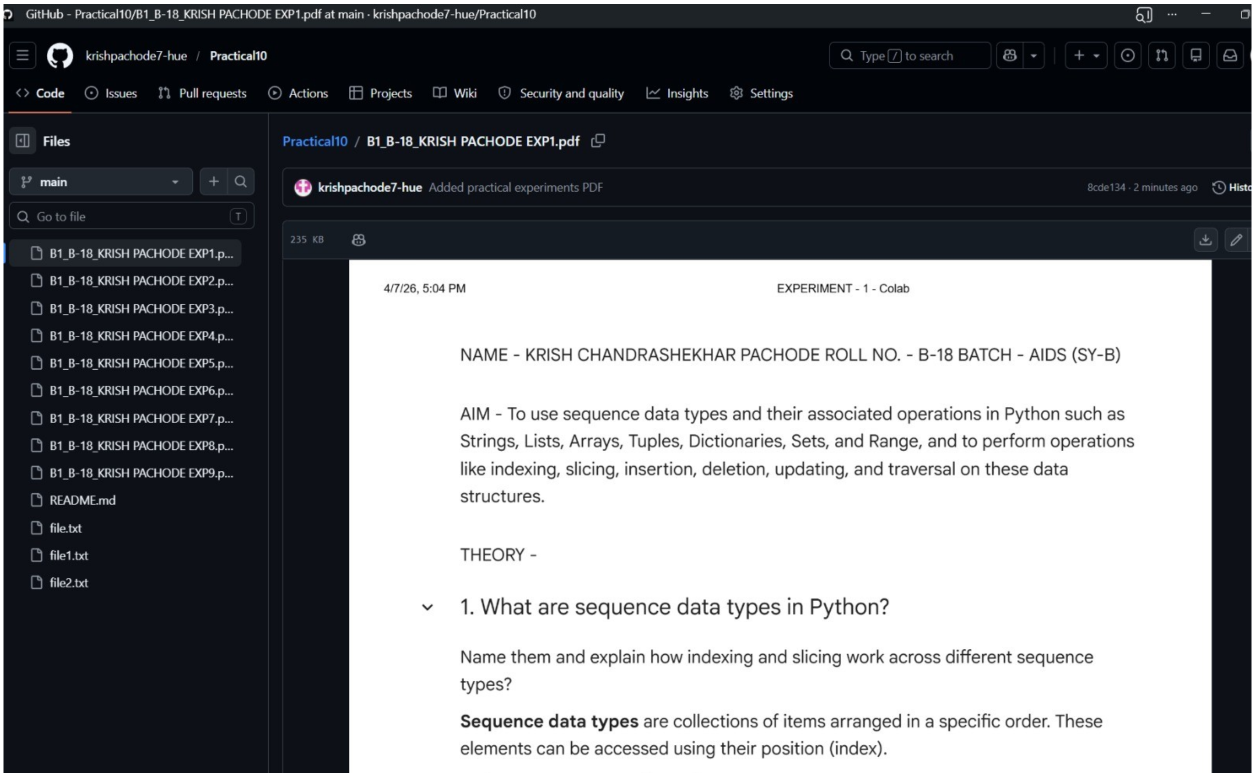
6. What is Version Control? Differentiate between Centralized vs Distributed VCS.

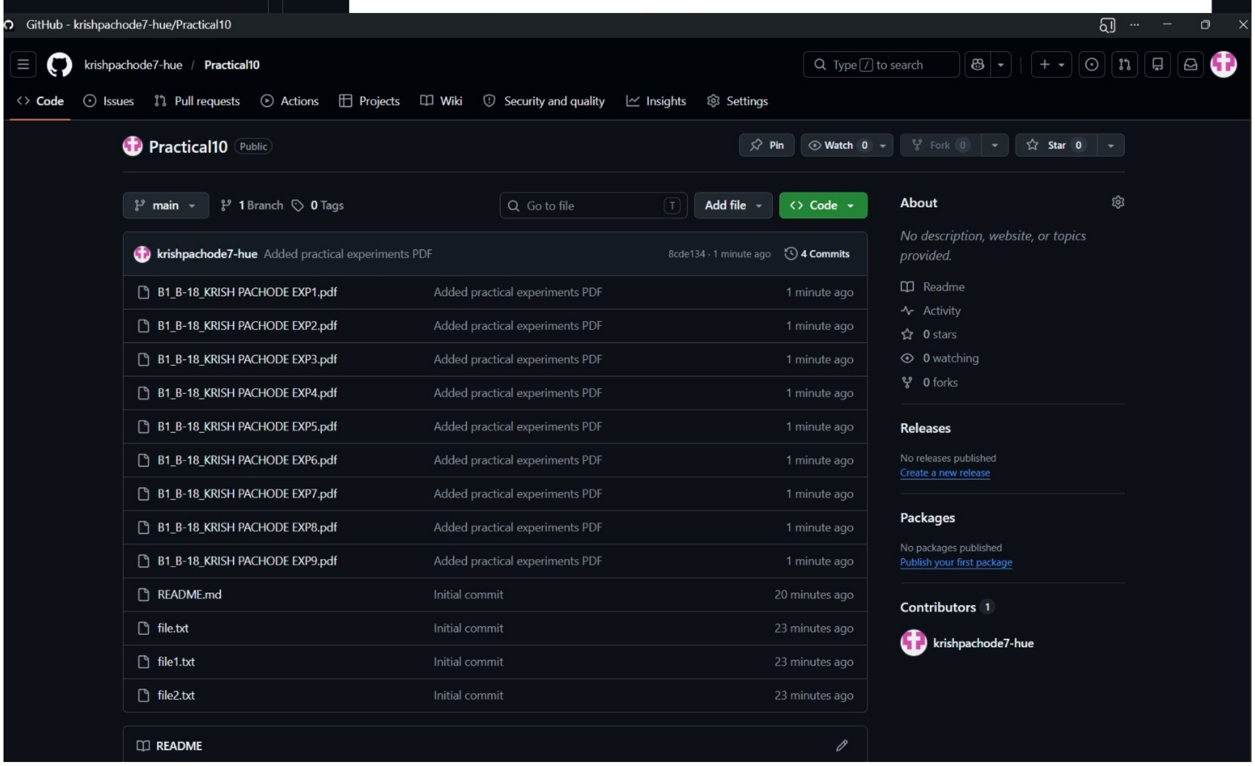
Version Control is a system that tracks changes in files and allows multiple users to collaborate efficiently.

Feature	Centralized VCS	Distributed VCS (Git)
Repository	Single central server	Every user has full copy
Internet Dependency	Required	Not always required
Speed	Slower	Faster
Example	SVN	Git

MINGW64:/c/Users/ASUS/Practical10

```
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 860 bytes | 95.00 KiB/s, done.
From https://github.com/krishpachode7-hue/Practical10
 * branch            main       -> FETCH_HEAD
 * [new branch]      main       -> origin/main
Merge made by the 'ort' strategy.
 README.md | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 README.md

ASUS@KRISH MINGW64 ~/Practical10 (main)
$ git push origin main
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 12 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (7/7), 620 bytes | 310.00 KiB/s, done.
Total 7 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), done.
To https://github.com/krishpachode7-hue/Practical10.git
   78b7bd6..1946b21  main -> main

ASUS@KRISH MINGW64 ~/Practical10 (main)
$ cd ~/Practical10

ASUS@KRISH MINGW64 ~/Practical10 (main)
$ ls
'B1_B-18_KRISH PACHODE EXP1.pdf'  'B1_B-18_KRISH PACHODE EXP6.pdf'   file.txt
'B1_B-18_KRISH PACHODE EXP2.pdf'  'B1_B-18_KRISH PACHODE EXP7.pdf'   file1.txt
'B1_B-18_KRISH PACHODE EXP3.pdf'  'B1_B-18_KRISH PACHODE EXP8.pdf'   file2.txt
'B1_B-18_KRISH PACHODE EXP4.pdf'  'B1_B-18_KRISH PACHODE EXP9.pdf'
'B1_B-18_KRISH PACHODE EXP5.pdf'   README.md

ASUS@KRISH MINGW64 ~/Practical10 (main)
$ git add .

ASUS@KRISH MINGW64 ~/Practical10 (main)
$ git commit -m "Added practical experiments PDF"
[main 8cde134] Added practical experiments PDF
 9 files changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 B1_B-18_KRISH PACHODE EXP1.pdf
 create mode 100644 B1_B-18_KRISH PACHODE EXP2.pdf
 create mode 100644 B1_B-18_KRISH PACHODE EXP3.pdf
 create mode 100644 B1_B-18_KRISH PACHODE EXP4.pdf
 create mode 100644 B1_B-18_KRISH PACHODE EXP5.pdf
 create mode 100644 B1_B-18_KRISH PACHODE EXP6.pdf
 create mode 100644 B1_B-18_KRISH PACHODE EXP7.pdf
 create mode 100644 B1_B-18_KRISH PACHODE EXP8.pdf
 create mode 100644 B1_B-18_KRISH PACHODE EXP9.pdf

ASUS@KRISH MINGW64 ~/Practical10 (main)
$ git push origin main
Enumerating objects: 12, done.
Counting objects: 100% (12/12), done.
Delta compression using up to 12 threads
Compressing objects: 100% (11/11), done.
Writing objects: 100% (11/11), 883.66 KiB | 27.61 MiB/s, done.
Total 11 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/krishpachode7-hue/Practical10.git
   1946b21..8cde134  main -> main
```


GitHub - Practical10/B1_B-18_KRISH PACHODE EXP1.pdf at main · krishpachode7-hue/Practical10
krishpachode7-hue / Practical10
Type / to search
Code
Issues
Pull requests
Actions
Projects
Wiki
Security and quality
Insights
Settings
Files
main
Go to file
B1_B-18_KRISH PACHODE EXP1.p...
B1_B-18_KRISH PACHODE EXP2.p...
B1_B-18_KRISH PACHODE EXP3.p...
B1_B-18_KRISH PACHODE EXP4.p...
B1_B-18_KRISH PACHODE EXP5.p...
B1_B-18_KRISH PACHODE EXP6.p...
B1_B-18_KRISH PACHODE EXP7.p...
B1_B-18_KRISH PACHODE EXP8.p...
B1_B-18_KRISH PACHODE EXP9.p...
README.md
file.txt
file1.txt
file2.txt
Practical10 / B1_B-18_KRISH PACHODE EXP1.pdf
krishpachode7-hue Added practical experiments PDF
8cde134 · 2 minutes ago
235 KB
4/7/26, 5:04 PM
EXPERIMENT - 1 - Colab
NAME - KRISH CHANDRASHEKHAR PACHODE ROLL NO. - B-18 BATCH - AIDS (SY-B)
AIM - To use sequence data types and their associated operations in Python such as Strings, Lists, Arrays, Tuples, Dictionaries, Sets, and Range, and to perform operations like indexing, slicing, insertion, deletion, updating, and traversal on these data structures.
THEORY -
1. What are sequence data types in Python?
Name them and explain how indexing and slicing work across different sequence types?
Sequence data types are collections of items arranged in a specific order. These elements can be accessed using their position (index).


GitHub - krishpachode7-hue/Practical10
krishpachode7-hue / Practical10
Type / to search
Code
Issues
Pull requests
Actions
Projects
Wiki
Security and quality
Insights
Settings
Practical10 Public
Pin
Watch 0
Fork 0
Star 0
main
1 Branch
0 Tags
Go to file
Add file
Code
krishpachode7-hue Added practical experiments PDF
8cde134 · 1 minute ago
4 Commits
B1_B-18_KRISH PACHODE EXP1.pdf Added practical experiments PDF 1 minute ago
B1_B-18_KRISH PACHODE EXP2.pdf Added practical experiments PDF 1 minute ago
B1_B-18_KRISH PACHODE EXP3.pdf Added practical experiments PDF 1 minute ago
B1_B-18_KRISH PACHODE EXP4.pdf Added practical experiments PDF 1 minute ago
B1_B-18_KRISH PACHODE EXP5.pdf Added practical experiments PDF 1 minute ago
B1_B-18_KRISH PACHODE EXP6.pdf Added practical experiments PDF 1 minute ago
B1_B-18_KRISH PACHODE EXP7.pdf Added practical experiments PDF 1 minute ago
B1_B-18_KRISH PACHODE EXP8.pdf Added practical experiments PDF 1 minute ago
B1_B-18_KRISH PACHODE EXP9.pdf Added practical experiments PDF 1 minute ago
README.md Initial commit 20 minutes ago
file.txt Initial commit 23 minutes ago
file1.txt Initial commit 23 minutes ago
file2.txt Initial commit 23 minutes ago
README
About
No description, website, or topics provided.
Readme
Activity
0 stars
0 watching
0 forks
Releases
No releases published
Create a new release
Packages
No packages published
Publish your first package
Contributors 1
krishpachode7-hue

Authentication Succeeded

You may now close this tab and return to the application.

```
$ mkdir Practical10

ASUS@KRISH MINGW64 ~
$ cd Practical10

ASUS@KRISH MINGW64 ~/Practical10
$ git init
Initialized empty Git repository in C:/Users/ASUS/Practical10/.git/

ASUS@KRISH MINGW64 ~/Practical10 (master)
$ echo

""

ASUS@KRISH MINGW64 ~/Practical10 (master)
$ echo "This is my  first file" > file.txt

ASUS@KRISH MINGW64 ~/Practical10 (master)
$ echo "Git Practical 10" >file2.txt

ASUS@KRISH MINGW64 ~/Practical10 (master)
$ 1s
bash: 1s: command not found

ASUS@KRISH MINGW64 ~/Practical10 (master)
$ echo "This is my first file" > file1.txt

ASUS@KRISH MINGW64 ~/Practical10 (master)
$ echo "Git Practical 10" > file2.txt

ASUS@KRISH MINGW64 ~/Practical10 (master)
$ 1s
bash: 1s: command not found

ASUS@KRISH MINGW64 ~/Practical10 (master)
$ ls
file.txt  file1.txt  file2.txt

ASUS@KRISH MINGW64 ~/Practical10 (master)
$ git add .
warning: in the working copy of 'file.txt', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'file1.txt', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'file2.txt', LF will be replaced by CRLF the next time Git touches it

ASUS@KRISH MINGW64 ~/Practical10 (master)
$ git commit -m "Initial commit"
[master (root-commit) 2d02a98] Initial commit
 3 files changed, 3 insertions(+)
 create mode 100644 file.txt
 create mode 100644 file1.txt
 create mode 100644 file2.txt

ASUS@KRISH MINGW64 ~/Practical10 (master)
$ git log
commit 2d02a9888b130f9797fde1d74ab840450e887b6c (HEAD -> master)
Author: Krish Pachode <krishpachode7@gmail.com>
Date:   Tue May 5 22:31:06 2026 +0530

    Initial commit

ASUS@KRISH MINGW64 ~/Practical10 (master)
$ |
```

Conclusion –

In this experiment, we understood the concepts of Git and GitHub along with important commands like git add, git commit, git push, and git pull. We also learned about staging area and version control systems, which are essential for managing and tracking changes in software development.